Lukas Folkman, PhD

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Education

2011–2015 **PhD Candidate**[1,2]
[1]Institute for Integrated and Intelligent Systems, Griffith University, Australia
[2]NICTA – Australia's ICT Research Centre of Excellence, Australia
thesis: Predicting stability and functional changes induced by protein mutations with a machine learning approach
advisers: Prof Yaoqi Zhou[1], Prof Bela Stantic[1], Prof Abdul Sattar[1,2]

2007–2010 **Student of Informatics (Master's Degree)**
Faculty of Informatics, Masaryk University, Czech Republic

2007 **Visiting Student**
School of Computer Science, University College Dublin, Ireland

2004–2007 **Student of Applied Informatics (Bachelor's Degree)**
Faculty of Informatics, Masaryk University, Czech Republic

Professional Experience

2018–now **Postdoctoral Researcher**
CeMM Research Center for Molecular Medicine, Austrian Academy of Sciences
adviser: Prof Christoph Bock (https://www.bocklab.org)

2016–2018 **Postdoctoral Researcher**
Department of Biosystems Science and Engineering, ETH Zurich, Switzerland
adviser: Prof Karsten Borgwardt (https://bsse.ethz.ch/mlcb)

2015–2016 **Visiting Researcher**
Institute for Integrated and Intelligent Systems, Griffith University, Australia
adviser: Prof Yaoqi Zhou (https://sparks-lab.org)

2008–2010 **Software Developer and Consultant**
ARTIN, Czech Republic

Funding

2019–now Marie Skłodowska-Curie Actions Individual Fellowship; *project name:* SingleCellAI

2011–2014 NICTA Project Award Scholarship

2011–2014 Griffith University Postgraduate Research Scholarship

2011–2014 Griffith University International Postgraduate Research Scholarship

Dec 2010 NICTA Summer Research Scholarship

Awards

Sep 2017 Online Poster Award (1st place) at [BC]^2 Basel Computational Biology Conference 2017 for *Kernelized rank learning for personalized drug recommendation*, Basel, Switzerland

Jul 2017 F1000Research Poster Award at ISMB/ECCB 2017 for *Kernelized rank learning for personalized drug recommendation*, Prague, Czech Republic

Jun 2014 Winner of Three-Minutes Thesis Competition, School of ICT, Griffith University

Apr 2014 Gold Coast Association of Postgraduates Conference Funding Award, Griffith University

Jan 2013 Griffith University Conference Travel Grant, Griffith University

Dec 2010 2nd best paper in the workshop Keymaker 2010, Brno, Czech Republic

May 2010 2nd place in Czech and Slovak university research competition, Ostrava, Czech Rep.

Mar 2010 Dean's award for outstanding thesis, Faculty of Informatics, Masaryk University

Mar 2010 Dean's award for excellent study results, Faculty of Informatics, Masaryk University

Research Interests

Applied machine learning
deep learning for single-cell transcriptomics
personalised drug recommendation systems
prediction of disease-causing and functionally-relevant genomic variants

Data science for the life sciences
integrative statistical analysis of high-dimensional datasets

First-Author Publications

These authors contributed equally

Moorlag# SJCFM, **Folkman# L**, ter Horst# R, Barreca# D, Krausgruber# T, Fife-Gernedl V, …, Netea MG, Bock C: Integrative analysis of factors explaining inter-individual heterogeneity in immune function, chromatin accessibility, and responses to BCG vaccination, *in preparation* for *Cell*.

He# X, **Folkman# L**, Borgwardt K: Kernelized rank learning for personalized drug recommendation, *Bioinformatics* (2018), 34(16), 2808–2816.

Livingstone# M, **Folkman# L**, Yang Y, Zhang P, Mort M, Cooper DN, Liu Y, Stantic B, Zhou Y: Investigating DNA-, RNA-, and protein-based features as a means to discriminate pathogenic synonymous variants, *Human Mutation* (2017), 38(10), 1336–1347.

Folkman L, Stantic B, Sattar A, Zhou Y: EASE-MM: sequence-based prediction of mutation-induced stability changes with feature-based multiple models, *Journal of Molecular Biology* (2016), 428(6), 1394–1405.

Folkman L, Yang Y, Li Z, Stantic B, Sattar B, Mort M, Cooper DN, Liu Y, Zhou Y: DDIG-in: detecting disease-causing genetic variations due to frameshifting indels and nonsense mutations by sequence and structural properties at nucleotide and protein levels, *Bioinformatics* (2015) 31(10), 1599–1606.

Folkman L, Stantic B, Sattar A: Feature-based multiple models improve classification of mutation-induced stability changes, *BMC Genomics* (2014) 15(Suppl 4), S6.

Folkman L, Stantic B, Sattar A: Towards sequence-based prediction of mutation-induced stability changes in unseen non-homologous proteins, *BMC Genomics* (2014) 15(Suppl 1), S4.

Folkman L, Stantic B, Sattar, A: Sequence-only evolutionary and predicted structural features for the prediction of stability changes in protein mutants, *BMC Bioinformatics* (2013) 14(Suppl 2), S6.

Co-Author Publications

de Bree LCJ, Mourits VP, Koeken VACM, Moorlag SJCFM, Janssen R, **Folkman L**, …, Netea MG: Circadian rhythm influences induction of trained immunity by BCG vaccination, *The Journal of Clinical Investigation* (2020), 130(10), 5603–5617.

Pejaver V, Babbi G, Casadio R, **Folkman L**, Katsonis P, Kundu K, …, Bromberg Y: Assessment of methods for predicting the effects of PTEN and TPMT protein variants, *Human Mutation* (2019), 40(9), 1495–1506.

Savojardo C, Petrosino M, Babbi G, Bovo S, Corbi-Verge C, Casadio R, Fariselli P, **Folkman L**, …, Capriotti E: Evaluating the predictions of the protein stability change upon single amino acid substitutions for the FXN CAGI5 challenge, *Human Mutation* (2019), 40(9), 1392–1399.

Clark WT, Kasak L, Bakolitsa C, Hu Z, Andreoletti G, Babbi G, Bromberg Y, Casadio R, Dunbrack R, **Folkman L**, …, LeBowitz JH: Assessment of predicted enzymatic activity of alpha-N-acetylglucosaminidase (NAGLU) variants of unknown significance for CAGI 2016, *Human Mutation* (2019) 40(9), 1519–1529.

Conference Proceedings

Folkman L, Pullan W, Stantic B: Generic parallel genetic algorithm framework for protein optimisation, *in* 'Algorithms and architectures for parallel processing', Vol. 7017 of *LNCS*, pp. 64–73, Springer, 2011.

Stetsko A, **Folkman L**, Matyas V: Neighbor-based intrusion detection in wireless sensor networks, *in* 'Wireless and Mobile Communications (ICWMC)', pp. 420–425, IEEE, 2010.

Oral Presentations

2014 Asia-Pacific Bioinformatics Conference, Vancouver, Canada
Towards sequence-based prediction of mutation-induced stability changes in unseen non-homologous proteins

2013 Conference on Intelligent Systems for Molecular Biology & European Conference on Computational Biology (ISMB/ECCB), SNP Special Interest Group, Berlin, Germany
Feature-based multiple models improve classification of mutation-induced stability changes

2013 Asia-Pacific Bioinformatics Conference, Vancouver, Canada
Sequence-only evolutionary and predicted structural features for the prediction of stability changes in protein mutants

2011 Symposium on Advances of Distributed Computing and Networking, Melbourne, Australia
Generic parallel genetic algorithm framework for protein optimisation

Poster Presentations

2018 German Conference on Bioinformatics, Vienna, Austria
Kernelized rank learning for personalized drug recommendation

2017 $[BC]^2$ Basel Computational Biology Conference, Basel, Switzerlan
Kernelized rank learning for personalized drug recommendation

2017 Conference on Intelligent Systems for Molecular Biology & European Conference on Computational Biology (ISMB/ECCB), Prague, Czech Republic
Kernelized rank learning for personalized drug recommendation

Teaching

2020 **Teaching Assistant**
CeMM Research Center for Molecular Medicine, Austrian Academy of Sciences
course: one-day machine-learning workshop for first-year PhD students

2016 **Teaching Assistant**
Department of Biosystems Science and Engineering, ETH Zurich, Switzerland
course: Data Mining I

2012–2013 **Teaching Assistant**
School of Information and Communication Technology, Griffith University, Australia
courses: Introduction to Programming, Object-Oriented Programming, Database Systems and Administration*****, Database Technology

*****Receving mean score of 4.78 (out of 5); with five (out of 12) nominations for an award: “Lukas was also an excellent tutor. He explains things so clearly and breaks up complex subjects into something simple which helped me understand topics better.”

Student Supervision

2017 Elli Tzini, Master’s thesis, ETH Zurich, Basel, Switzerland

2016 Caroline Weis, Master’s thesis, ETH Zurich, Basel, Switzerland

2013 Mark Livingstone, Honours thesis, Griffith University, Southport, Australia

Institutional Service

2020–now Data ambassador and manager, part of a working group for implementing institute-wide “FAIR” principles and transfering 650 TB of molecular biology data to new storage

2012–2013 Founding chair of the IIIS/ICT Higher-Degree-Research Student Community

Reviewing

Bioinformatics, Journal of Machine Learning Research, Journal of Molecular Biology, Scientific Reports, European Conference on Machine Learning

Skills

Languages English – advanced (IELTS band score 8), German – beginner, Slovak – native

Coding Python – advanced, Java, C, C++, R, SQL, Linux shell, parallel programming with MPI